DAY 18 – MACHINE TEST (SSH & SSL/TLS)

1. Check SSH availability: Run 'ssh' and write the output.

2. Verify GitHub SSH authentication: Run 'ssh -T git@github.com' and write the output.

3. List SSH files: Run 'ls -la ~/.ssh' and identify the Public Key and Private Key.

4. Generate a new SSH Key using 'ssh-keygen -t ed25519 -C "your_email@gmail.com"' and verify the files are created.

5. Start SSH Agent using 'eval "$(ssh-agent -s)"' and write the output.

6. Add SSH Key to Agent using 'ssh-add ~/.ssh/id_ed25519' and write the output.

7. Display Public Key using 'cat ~/.ssh/id_ed25519.pub'. Copy the first few characters of the key.

8. Check HTTPS Response using 'curl -I https://google.com'. What status code is returned?

9. Compare HTTP and HTTPS using 'curl -I http://google.com' and 'curl -I https://google.com'. Write two differences.

10. Scenario Task: AWS EC2 IP = 54.234.xxx.xxx, key file = mykey.pem. Write the SSH command used to connect.

11. What is SSH?

12. Why is SSH required in DevOps?

13. Which port does SSH use?

14. What is SSH Key Authentication?

15. Difference between Public Key and Private Key.

16. What is SSL?

17. What is TLS?

18. Difference between SSL and TLS.

19. Why is HTTPS more secure than HTTP?

20. Which port does HTTPS use?

21. What is an SSL/TLS Certificate?

22. What information is stored inside a certificate?

23. What does the 'Connection is Secure' message in a browser indicate?

24. Explain the SSL/TLS Handshake flow: Client Hello, Server Hello, Certificate, Verification, Key Exchange, Encrypted Session.

25. Explain the production flow: User → DNS → HTTPS → Load Balancer → Nginx → Docker Container → Database → Response.

26. Explain the output of 'curl -I https://google.com'.

27. Compare the outputs of HTTP and HTTPS requests.

28. List three real-world uses of SSH in DevOps.

29. List three real-world uses of SSL/TLS in modern applications.

30. Explain the complete process of connecting to an AWS EC2 instance securely using SSH keys.